CS215 OPEN PROJECT · GROUP 02

The Perks of Being a First Mover

How many rating points is White’s first move worth on Lichess?

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THE QUESTION

Three hypotheses about White’s edge

H1

Rating

The advantage peaks at intermediate ratings and shrinks at 2200+, where Black knows how to neutralise it.

H2

Openings

Aggressive gambits give White a bigger edge than drawish openings, which also end in more draws.

H3

Time control

With more time on the clock, White uses the initiative better, so the advantage grows from Bullet to Classical.

DATA

Every rated Lichess game, Jan–Feb 2024

190.6M

games read from the Lichess Open Database

162.2M

games used: finished, with a rating gap of at most 100

60+ GB

of compressed data, streamed from Lichess and never saved to disk

From each game we read only the headers: the result, both ratings, the time control and the opening. The games are counted by opening, time control, 100-point rating bracket and rating gap, in **counts.csv**: a file of a few MB from which every result can be rebuilt in seconds.

METHOD

White plays like someone h points stronger

1 · Model

$$E = 1 / (1 + 10^{(-(d + h)/400)})$$

White's Elo expected score when rated d points above Black, with a bonus of h rating points for moving first.

2 · Maximum likelihood

$\hat{h}$ is the value at which White's actual total score equals the expected total, found with Newton's method.

3 · Confidence interval

From the Fisher information I: SE = $1/\sqrt{I}$, and the 95% CI is $\hat{h} \pm 1.96 \cdot SE$.

4 · Comparisons

Two groups of games are compared with a z-test: $z = (\hat{h}_1 - \hat{h}_2) / \sqrt{(SE_1^2 + SE_2^2)}$.

METHOD · LOG-LIKELIHOOD

Choosing the h that best explains the games

1	$P(\text{game } i) = p_i^{s_i} \cdot (1 - p_i)^{(1 - s_i)}$	s_i = White's actual score (1, ½ or 0); p_i = its predicted score from the model
2	$\ell(h) = \Sigma\, [\, s_i \ln p_i + (1 - s_i) \ln(1 - p_i)\,]$	Games are independent; the log turns the product of 162M probabilities into a sum
3	$\ell'(h) = K \Sigma\, (s_i - p_i) = 0 \Rightarrow \Sigma\, s_i = \Sigma\, p_i(\hat{h})$	At $\hat{h}$, White's actual total score equals the predicted total
4	$\ell''(h) = -K^2 \Sigma\, p_i(1 - p_i) < 0$ $h \leftarrow h + \Sigma(s_i - p_i) \,/\, (K \Sigma\, p_i(1 - p_i))$	One peak, no local maxima; Newton's method reaches it in 3–5 steps
5	$I = -\ell''(\hat{h})$, $SE = 1/\sqrt{I}$, 95% CI $= \hat{h} \pm 1.96 \cdot SE$	The sharper the peak, the more precisely h is known

$p_i = 1 / (1 + 10^{(-(d_i + h)/400)})$, $K = \ln 10 / 400$. A draw counts as $s = ½$. All games are pooled: one game alone would give $h = \pm\infty$.

OVERALL RESULT · 162 MILLION GAMES

+11.9 points

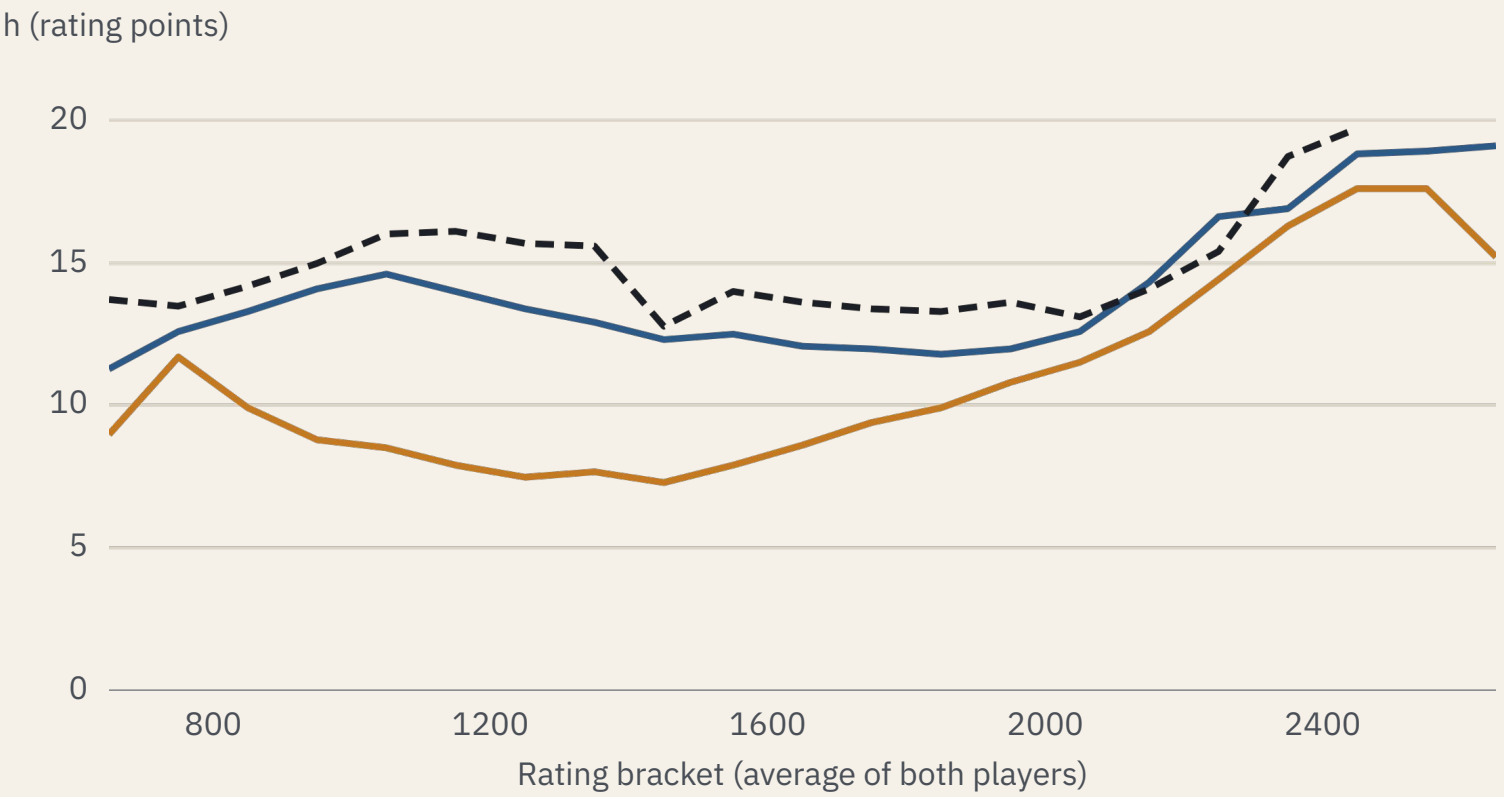
White plays like someone 11.9 rating points stronger (95% CI ± 0.1).

Against an equally rated opponent, White scores **51.7%** instead of 50%.

All games: White wins 49.7% · Draws 4.0% · Black wins 46.3%

H1 · RATING

The edge grows at the top, not the middle

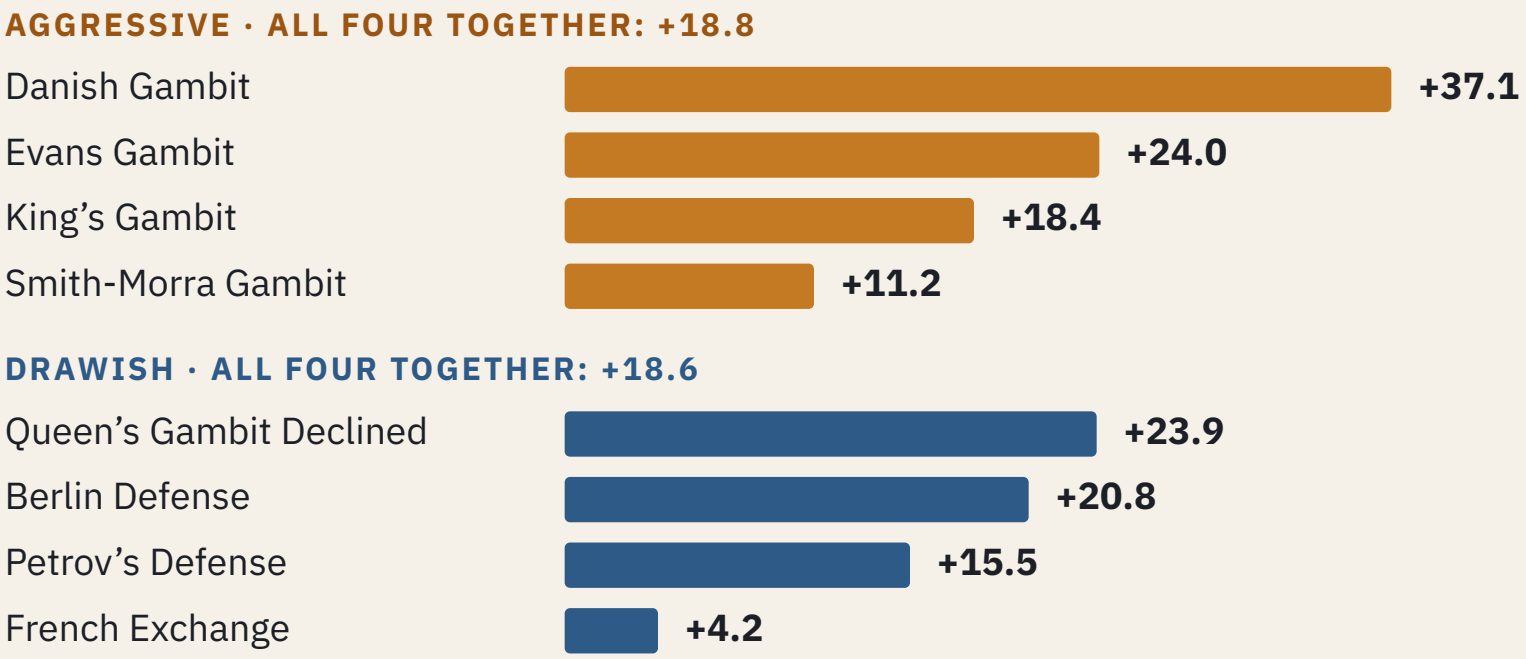


There is no peak in the middle. In Bullet, h falls to about **7** near 1400 and climbs to **18** at 2400+. Blitz and Rapid stay at 12–16 in the middle and reach **19–20** at the top.

H1 rejected: strong players get more out of the first move, not less.

H2 · OPENINGS

The opening matters; its “style” doesn’t



+0.2 ± 0.4

Aggressive minus drawish, all speeds: z = 1.0, p = 0.3. No real difference.

Yet single openings range from **4** to **37** points.

H2 not supported, although drawish openings do end in more draws: 4.4% vs 3.2%.

H2 · BY TIME CONTROL

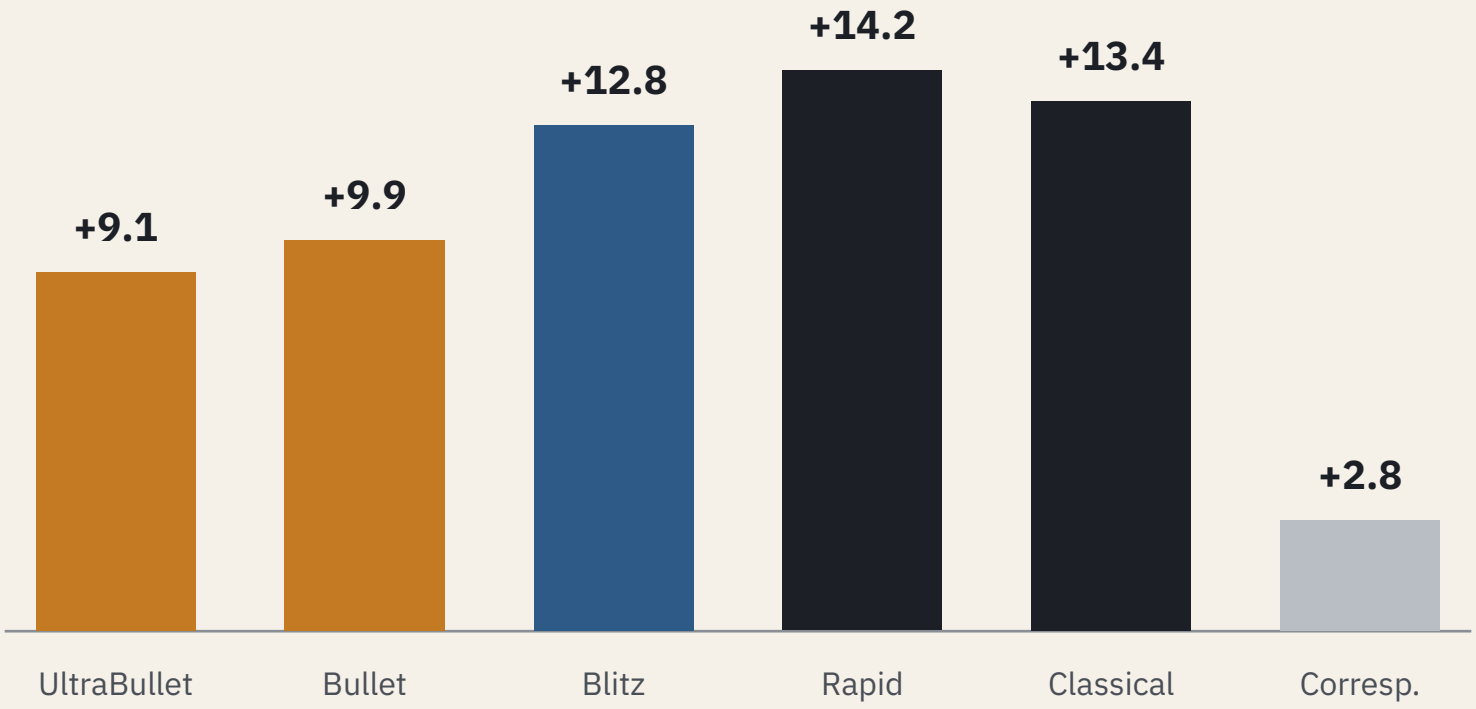
The gap stays within 2 points at every speed

Speed	Aggressive h	Drawish h	Difference (95% CI)	p
All speeds	+18.8	+18.6	+0.2 ± 0.4	0.3
Bullet	+16.9	+16.3	+0.6 ± 0.8	0.10
Blitz	+19.5	+18.9	+0.6 ± 0.6	0.04
Rapid	+20.3	+22.4	−2.1 ± 1.1	0.0002

With millions of games even tiny gaps can be “significant”, so we look at their size: never more than about 2 rating points, and in Rapid the **drawish openings are slightly ahead**.

H3 · TIME CONTROL

More time helps White, up to Rapid



Blitz vs Bullet: **+2.9** (z = 48)

Rapid vs Blitz: **+1.4** (z = 17)

Classical vs Rapid: **−0.8** (p = 0.04)

Correspondence (grey bar) has only 81k games, so its interval is ± 2.4.

H3 partly supported: h rises from Bullet to Rapid, then levels off.

CHALLENGES

What made it hard, and how we handled it

Tens of GB per month

Each month is streamed from Lichess and decompressed on the fly by our C++ program on Colab; only the counts are saved.

Messy opening names

We checked all 3,096 Lichess opening names (in 2024 the Petrov was still the “Russian Game”) and excluded gambit sidelines.

Everything is “significant”

We report effect sizes in rating points with 95% CIs; an independent re-fit in pandas reproduces the C++ results.

Odd data at 2900+

A few top brackets have 50–96% draws, probably bots or arranged games, so we left them out of the rating charts.

CONCLUSION

Moving first is worth about 12 points

H1	**Rejected.** The edge is smallest at middle ratings and largest at 2200+, up to about 19 points.
H2	**Not supported.** Aggressive ≈ drawish (+18.8 vs +18.6), but single openings range from 4 to 37 points.
H3	**Partly supported.** h rises from Bullet (9.9) to Rapid (14.2), then levels off.

C++17 + zstd · Google Colab · pandas / NumPy · LaTeX + pgfplots · Plotly dashboard